Venkata Lakshmi Parimala Pasupuleti

Staff Software Engineer · Python · RAG Pipelines · LLM Evaluation · Document Processing · Agentic AI · AWS · FastAPI · Production AI Systems

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McLean area, Virginia — Remote

PROFESSIONAL SUMMARY

Software engineer with 8 years shipping production-grade systems and a focused track record in production document AI — the exact stack JunieAI is built on. Built **Leasel**: end-to-end document processing pipeline over 1,240 complex legal PDFs (chunking, embedding, retrieval, deterministic rules engine) achieving **93.6% precision** with full auditability on high-stakes outputs. Built **AgentProbe**: 3-agent LangGraph runtime with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance), pass@k regression gates, and HITL feedback loops — blocking failing runs before production. At Entrata: **20+ modules end-to-end** (1M+ records, 100K+ MAU, 99.5% uptime) including the first production AI pipeline on **1.6M real financial records** with quality gates and output validation; led Docker microservices rollout (50% speedup) and AWS migration (100% PHI compliance). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Production AI · Document Pipelines · AWS · Microservices · Cross-functional Delivery

- Owned **20+ production-grade modules end-to-end** (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints — **1M+ records, 100K+ MAU, 99.5% uptime**; led architecture, design patterns, implementation, rollout, and adoption tracking for every feature shipped.
- Shipped Entrata's **first production AI pipeline on 1.6M real financial records** (PyTorch, OpenAI API); implemented output validation, quality gates, and human-in-the-loop review before customer exposure; led Docker microservices rollout (50% reliability gain); migrated to **AWS S3/RDS/Glue** (100% PHI compliance, zero incidents); optimized 200+ SQL queries (30% performance improvement).
- Built **HUD and LIHTC (Section 42) compliance tooling** — income eligibility, AMI thresholds, Section 8/HCV certification, tax credit recertification pipelines across 100K+ unit portfolios; eliminated **80% of manual workload** with deterministic rules engine and full audit trails; zero compliance violations.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Data Pipelines · API Integrations · Compliance Automation

- Built compliance automation processing **millions of records** — income limits, household composition rules, asset verification thresholds encoded into deterministic systems with full audit trails; built **API integrations and webhooks** across property management systems; conducted 300+ peer code reviews.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; rapid prototyping of new reporting tooling from 0→1 through production rollout across 100K+ unit portfolios.

PROJECTS

Leasel: Production Document AI Pipeline
DOCUMENT PROCESSING · RAG · LLM EVAL

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 complex legal PDFs — end-to-end document processing pipeline: extraction, chunking, embedding, vector retrieval, deterministic rules engine for fully auditable high-stakes outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring and regression gates.

AgentProbe: Multi-Agent AI Eval Framework
MULTI-AGENT · LLM-AS-JUDGE

Python · LangGraph · FastAPI · ChromaDB · pgvector-compatible · OpenAI · Anthropic · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling and inter-agent state handoffs; **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with HITL feedback loops and CI/CD regression gates — blocking failing runs before production; 15-endpoint FastAPI backend, React observability dashboard.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
HEALTHCARE

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, R, PHP; FastAPI, Django, Flask, REST APIs, webhooks, API integrations, RabbitMQ, event-driven distributed systems

AI & Document Processing: LangGraph, LangChain, Agentic AI, RAG Pipelines, Document Extraction, Chunking & Embedding, Vector Stores (ChromaDB, pgvector, Pinecone), LLM-as-Judge, LLM Evaluation, HITL Feedback Loops, Pass@k Eval, Multi-Agent Orchestration, Tool Calling, OpenAI API, Anthropic API, PyTorch, BERT, NLP, FHIR

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, Microservices, CI/CD, GitHub Actions, PostgreSQL, MySQL, system design, scalable architecture, design patterns, HIPAA, PHI compliance, Git, Linux